

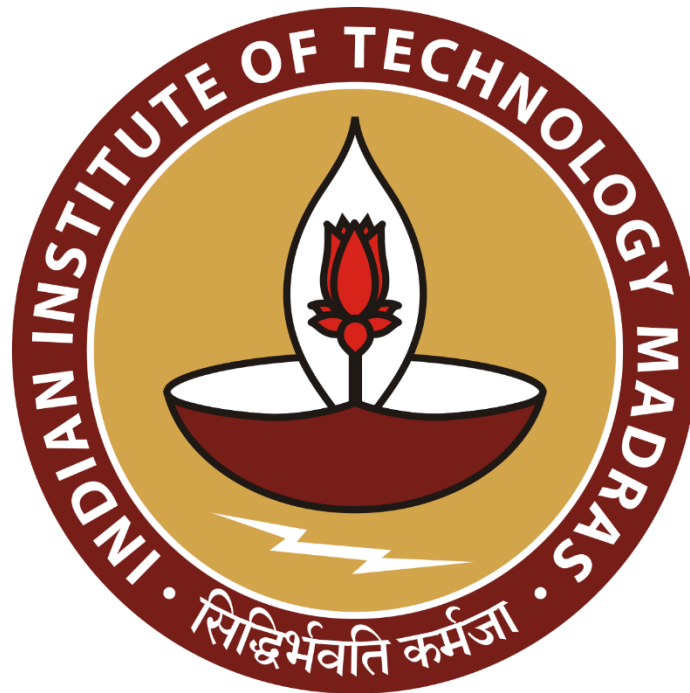
The Seasonal Sales and Demand Trends In Sahitya Sadan Bookstore

A Mid Term report for the BDM capstone Project

Submitted by

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1. Executive Summary

The Sahitya Sadan bookstore, located in the Musabani market, East Singhbhum, Jharkhand is a small B2C business established by Mr. Durga Shankar's father back in 1972. Through discussion with the owner, it has been identified that the business is facing an Inventory Management Issues (overstocking of low demand items and stockouts for the high demand items on a particular period of time).

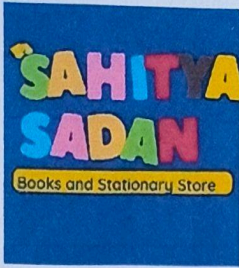
The project focuses on analyzing the 3-months of sales data, the Sahitya Sadan Bookstore uses the "My BillBook" application in which each and every invoice's data is maintained. The data received from the business was raw and contained so many columns which was not necessary for the data analysis, hence it was removed.

The data analysis process was started just after cleaning the raw data. Since, the business is facing the Inventory Management issues, to address the same and to increase the profits of the business, the Descriptive analysis was done first in order to understand the fundamental characteristics of the data, after this the ABC analysis was done which correctly identified the products that are financially critical to the business and divided the items into three categories, which will help the business to carefully manage the inventory. After this the Trends analysis was done in order to better understand and identify the patterns in the data. Through all the above mentioned analysis, one most important insight came out that "Books are the primary revenue driver".

2. Proof of Data Originality

Details of the Business :

- Shop Name : Sahitya Sadan
- Owner name : Mr. Durga Shankar
- Number of Employee : 1 (Mr. Jagmohan Patar)
- Type of Organisation : B2C (Business-to-Consumer)
- Mobile no. : 9204091062
- Address : Musabani market, East Singhbhum, Jharkhand, 832104
- Bookstore picture-1 : [Image_1](#)
- Bookstore picture-2 : [Image_2](#)
- Bookstore picture-3 : [Image_3](#)
- Bookstore picture-4 : [Image_4](#)
- Bookstore picture-5 : [Image_5](#)
- Interaction Video Link : [Interaction_Video](#)
- English transcript PDF for the video : [English_Transcript](#)



**SAHITYA
SADAN**

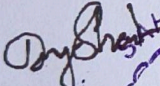
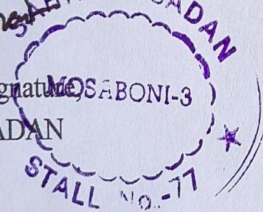
Musaboni Market, Shop no 77,
Musabani, Jharkhand 832104, India
Mobile : 9204091062
GSTIN : 20COUPS4093B1ZT
PAN Number : COUPS4093B

03/11/2025

TO WHOMEVER IT MAY CONCERN

This is to confirm that Mr. Aman Kumar Mahali has collected data from our company for the purpose of his BDM Capstone Project.

By this letter, we provide our consent for utilization of data for the above-mentioned project activities at Indian Institute of Technology, Madras. We hereby issue this No Objection Certificate on receiving confirmation from Mr. Aman Kumar Mahali that the data is intended for IITM internal use only and will be disguised appropriately when disclosed for public purposes.


Authorised (Signature) 
SAHITYA SADAN

Image_6 : Authorization Letter

3. Meta Data

The sales data of 3 months, from 1st April, 2024 to 30th June, 2024 was collected from the bookstore. The Sahitya Sadan bookstore uses the application named “My BillBook”, and all the sales data are being stored in the application only. Initially the data was received in a raw format which contains the following columns : **Invoice No, Invoice Date, Contact Name, Amount, Remaining Amount, Invoice Status, Due Date, Invoice Link, Payment Type, Created by**, etc. Then each and every invoice’s data was extracted into another Google Sheet and a pivot table was created. Since, the business is facing Inventory Management Issues, to better deal with it an additional columns of “**Serial No.**” and “**Category**” has been added which will help to better analyse the sales data and to solve the issue. The sheet has the following columns and the picture has been attached below :

1. **Invoice Date** : This column contains the date on which the invoice was generated.
2. **Serial No.** : This column contains the sequential numbering for each item.
3. **Items** : This column contains the name of the items being sold.
4. **Quantity** : This column contains the number of pieces being sold for each item.
5. **MRP** : This column contains the Maximum Retail Price of each item.
6. **Total** : This column contains the total bill, by multiplying the MRP and the Quantity.
7. **Category** : This column contains the categories of the items like Books, Stationary, Paper Goods and Art materials.

Raw Dataset Link : [Raw Dataset](#)

Cleaned Dataset Link : [Cleaned Dataset](#)

Invoice Date	Serial No.	Items	Quantity	MRP	Total	Category
01/04/2024	1	Spiral ESS Eight Notebook	1 PCS	15	15	Paper Goods
01/04/2024	2	Single Brown Sheet	6 PCS	7	42	Paper Goods
01/04/2024	3	Nalanda Sports Long Exercise Book 440 Pages	3 PCS	150	420	Paper Goods
01/04/2024	4	Saathi Long 224 Pages Rule Copy	1 PCS	105	105	Paper Goods
01/04/2024	5	Classmate 140 Pages Exercise Book (29.7 × 21 cm)	1 PCS	65	65	Paper Goods
01/04/2024	6	Saathi Long 160 Pages Rule Copy	2 PCS	70	140	Paper Goods
01/04/2024	7	Vidya Bharti Square Copy 120 Pages	1 PCS	30	30	Paper Goods
01/04/2024	8	Amar Drowning Book 32 Pages	1 PCS	30	30	Art materials
01/04/2024	9	Prachi Mathematics 1	1 PCS	259	259	Books
01/04/2024	10	SM Kala Pujan 2	1 PCS	53	53	Books
01/04/2024	11	Hindi Sulekh Mala 2	1 PCS	42	42	Books
01/04/2024	12	Envelope Blue Cloth Line 41.5 * 31 cm	1 PCS	15	15	Paper Goods
01/04/2024	13	Camlin Fine Tip Permanent Marker	1 PCS	10	10	Stationary
01/04/2024	14	Transparent Cello Tape 2 Inc	1 PCS	60	60	Stationary
01/04/2024	15	Rule Quire Paper No 6	1 PCS	30	30	Paper Goods
01/04/2024	16	Loose Use And Throw Blue Pens	20 PCS	4	80	Stationary
01/04/2024	17	Nalanda Sports 176 Pages Rule Copy	2 PCS	60	120	Paper Goods

4. Descriptive Statistics

4.1 Measures of Central Tendency

4.1.1 Mean : This measure of central tendency gives the average sales amount for the different categories.

For Stationary category : ₹29.56

For Books Category : ₹143.67

For Paper Goods Category : ₹91.66

For Art materials Category : ₹44.06

The mean value for the “**Stationary**” category (₹29.56) signifies that the cheap items (pens, pencils, and other stationaries) that are almost always bought in multiples.

The mean value for the “**Books**” category (₹143.67) signifies that the books are the primary revenue driver and a single book transaction brings in, on average, 3 to 5 times more money than any other type of item.

The mean value for the “**Paper Goods**” category (₹91.66) signifies that the Paper Goods are the secondary revenue driver and customers spend a significant amount here, likely by buying either large, expensive items (like 400-page registers) or multiple smaller notebooks at once.

The mean value for the “**Art materials**” category (₹44.06) signifies that the customers buy these items less frequently, and the items themselves have a modest price, but it's important for being a "one-stop shop" for students.

4.1.2 Median (Quantity) : This measure of central tendency gives the median quantity sold.

For Stationary category : 1.0

For Books Category : 1.0

For Paper Goods Category : 1.0

For Art materials Category : 1.0

A median of **1.0** for every single category sends a very strong and clear message that at least 50% of all your transaction lines are for a single item (1 PCS).

4.1.3 Mode (Category) : This measure of central tendency gives the category which is appearing most often for a given month.

For April, 2024 : Books Category

For May, 2024 : Books Category

For June, 2024 : Paper Goods

In April 2024, **Books** were overwhelmingly the most frequent item sold (1,455 transactions) due to the "back-to-school" rush. This aligns perfectly with the previous finding that "Books" drove the most revenue. In this case, they were both the highest value and the most frequently sold item.

In May 2024, **Books** remained the most frequent sale, but the volume dropped dramatically (from 1,455 to 495 transactions). This shows the "back-to-school" rush was over, but some customers were still buying books.

In June 2024, **Paper Goods** (445 transactions) overtook **Books** (306 transactions) as the most frequent sale. This signals a change in customer purchasing behavior as the academic term settled. The major, one-time book purchases were finished, and now the more common, recurring need was for notebooks and other paper goods.

4.2 Measures of Variability

4.2.1 Standard deviation (SD) : This measure of variability is a measure of how dispersed the data is in relation to the mean.

For Stationary category : ₹43.03

For Books Category : ₹96.47

For Paper Goods Category : ₹103.41

For Art materials Category : ₹30.21

The SD value for the “Stationary” category signifies that number is much lower but not zero, it means that most stationary items are low-priced (like ₹5 erasers), but there are enough "premium" items (like DOMS Instrument box costs ₹100), to create a moderate spread.

The SD value for the “Books” category is very high, confirming the earlier insight. The store sells a vast range of books, from BP Social Science (₹25) to Prachi Mathematics-1 (₹259). This high spread is a sign of a healthy, diverse book inventory.

The highest SD value for the “Paper Good” category signifies that it has the widest price range from Single Brown Sheet (₹7) to Nalanda Long Exercise Book (₹420)

The lowest SD value for the “Stationary” category signifies that the total sale price for the art materials is the most predictable. Almost all items in this category fall into a tight, low-price band. There are very few, if any, expensive "premium" art supplies.

4.2.2 Range : This measure of variability is the difference between the maximum and the minimum sale price in the category.

For Stationary category : ₹567

For Books Category : ₹1,636

For Paper Goods Category : ₹838

For Art materials Category : ₹170

The **Books** category has the widest range, it means that it has the single most expensive item (at ₹1,641) and also some very cheap items (at ₹5), confirming it has a huge and diverse inventory.

The **Art materials** category has the narrowest range, confirming that all items in this category are low-priced and there are no expensive, premium products.

5. Explanation of Analysis Process/Method

As mentioned in the above section, 3 months (April 2024-June 2024) of data has been collected from the Sahitya Sadan bookstore, then each and every invoice's data was extracted into another Google Sheets and pivot table was created, For the better analysis of the data, the un-necessary columns like Invoice No., Contact Name, Invoice Status, Invoice link, etc. has been removed and an additional column of “**Serial No.**” and “**Category**” has been added which helped to categorize the items being sold based on Stationaries (like pen, pencil, fevicol, files, cello tapes, etc.), Books (like NCERT, JCERT, guide books, G.K., dictionary, etc.), Paper Goods (like Notebooks, registers, forms, diary, maps, etc.) and Art materials (like colors, drawing copy, brush, coloring books, etc.). Since the business is facing an Inventory Management Issues, it's necessary to categorize the items for the better analysis to solve the issue.

The category column was created using the function shown in the below picture :

Paper Goods <	Category
=IF(OR(ISNUMBER(SEARCH("Pen",C2)), ISNUMBER(SEARCH("Pencil",C2)), ISNUMBER(SEARCH("Permanent Marker",C2)), ISNUMBER(SEARCH("Cello Tape",C2)), ISNUMBER(SEARCH("Loose Use And Throw Blue Pens",C2)), ISNUMBER(SEARCH("Pencils For Rs 5",C2)), ISNUMBER(SEARCH("Eraser(Rubber)",C2)), ISNUMBER(SEARCH("Stapler",C2)), ISNUMBER(SEARCH("Scissors",C2)), ISNUMBER(SEARCH("Pencil For Rs-6",C2)), ISNUMBER(SEARCH("marker",C2)), ISNUMBER(SEARCH("Exam Board",C2)), ISNUMBER(SEARCH("Scale",C2)), ISNUMBER(SEARCH("Inc",C2)), ISNUMBER(SEARCH("PREMIER CUTTER",C2)), ISNUMBER(SEARCH("Fevicol",C2)), ISNUMBER(SEARCH("INSTRUMENT BOX",C2)), ISNUMBER(SEARCH("Slate",C2)), ISNUMBER(SEARCH("Chalk",C2)), ISNUMBER(SEARCH("File",C2)), ISNUMBER(SEARCH("Thermocol",C2)), ISNUMBER(SEARCH("highlighter",C2))), "Stationary", IF(OR(ISNUMBER(SEARCH("Copy",C2)), ISNUMBER(SEARCH("Forms",C2)), ISNUMBER(SEARCH("Spiral ESS Eight Notebook",C2)), ISNUMBER(SEARCH("Single Brown Sheet",C2)), ISNUMBER(SEARCH("Sports Long Exercise Book",C2)), ISNUMBER(SEARCH("Long Pages Rule Copy",C2)), ISNUMBER(SEARCH("Exercise Book",C2)), ISNUMBER(SEARCH("Envelope",C2)), ISNUMBER(SEARCH("Rule Quire Paper No 6",C2)), ISNUMBER(SEARCH("SM English Writing Copy",C2)), ISNUMBER(SEARCH("Notebook",C2)), ISNUMBER(SEARCH("Register",C2)), ISNUMBER(SEARCH("Stickers",C2)), ISNUMBER(SEARCH("Book Cover",C2)), ISNUMBER(SEARCH("nalanda bound register",C2)), ISNUMBER(SEARCH("Page Rule Copy",C2)), ISNUMBER(SEARCH("A4 Size Copy",C2)), ISNUMBER(SEARCH("A4 Sheet",C2)), ISNUMBER(SEARCH("Practical Notebook",C2)), ISNUMBER(SEARCH("Amar Deluxe 120 Page F/Rule",C2)), ISNUMBER(SEARCH("Super Choice Book Cover",C2)), ISNUMBER(SEARCH("Calender",C2)), ISNUMBER(SEARCH("Map",C2)), ISNUMBER(SEARCH("Vidya Bharti Square Copy120 Pages",C2))), "Paper Goods", IF(OR(ISNUMBER(SEARCH("Prachi Mathematics",C2)), ISNUMBER(SEARCH("English Conversation",C2)), ISNUMBER(SEARCH("Vandana",C2)), ISNUMBER(SEARCH("SM",C2)), ISNUMBER(SEARCH("Beni Madhav Shil Panjika Bangla",C2)), ISNUMBER(SEARCH("BHOLANATH ALL IN ONE",C2)), ISNUMBER(SEARCH("Apna Vyakaran",C2)), ISNUMBER(SEARCH("guide Book",C2)), ISNUMBER(SEARCH("NCERT",C2)), ISNUMBER(SEARCH("JCERT",C2)), ISNUMBER(SEARCH("Dictionary",C2)), ISNUMBER(SEARCH("Grammar",C2)), ISNUMBER(SEARCH("Hindi Sulekh Mala 2",C2))), "Books", IF(OR(ISNUMBER(SEARCH("point",C2)), ISNUMBER(SEARCH("color",C2)), ISNUMBER(SEARCH("brush",C2)), ISNUMBER(SEARCH("Drowning Book",C2)), ISNUMBER(SEARCH("brush",C2)), ISNUMBER(SEARCH("decorative paper",C2))), "Art materials"))))	

The analysis process begins just after cleaning and organizing the raw data. As mentioned in the problem statement that the bookstore is facing the Inventory Management Issue (stockouts for the high-demand items and overstocking of low-demand items), and to solve the same, a three-stage analytical process has to be done. This process moves logically from a broad understanding of the data (Descriptive Analysis) to a prioritized, then item-level classification (ABC Analysis) and finally a Trend Analysis.

5.1 Descriptive Analysis :

The descriptive analysis is done in order to understand the fundamental characteristics of the 3-months sales data, because it summarizes 4,983 individual transactions into a clear, interpretable story. This involved the calculations of key statistical measures of central tendency (Mean, Median, Mode) and the measure of variability (Standard deviation, Range).

- The **Mean (Total)** for "Books" (₹143.67) immediately flagged it as a high-value category.
- The **Median (Quantity)** which came out 1.0 for all categories revealed that most transactions are for single items, highlighting the outlier importance of bulk purchases.
- The **Mode (Category)** shifting from "Books" in April to "Paper Goods" in June was the first clear signal of a changing demand pattern.
- The **high Standard Deviation** for "Paper Goods" (₹103.41) proved its price inconsistency (mix of cheap and expensive items), justifying a more careful inventory approach.
- The **Range** for "Books" (₹1,636) category confirmed that it has the widest range and diverse inventory.

Hence, the descriptive analysis helped to confirm that "Books" and "Paper Goods" are the high demand items in April, which could lead to stock-out, and "Stationary" are high-volume but low demand (in terms of revenue) items which could easily lead to over-stocking, during this month.

5.2 ABC Analysis :

The ABC analysis is done to classify every unique item in the inventory. First, all 4,983 transactions are aggregated by the 347 unique Items name to find the total revenue generated by each specific product then all items are sorted in descending order by their total revenue

- **Category A (76 items)** : This category includes the high-demand products, or items which are less in number but make up almost 80% of total revenue.

Eg : JCERT(Jharkhand Council of Educational Research and Training) Books from primary to Secondary classes, Nalanda Long Exercise Books, etc.

- **Category B (99 items)** : This category includes mid-range products, or items which are more in number than Category A and make up almost 15% of the total revenue.

Eg : Amar Drawing Books, Doms Instruments Box, Saathi A4size Notebook, etc.

- **Category C (172 items)** : This category includes low-demand products, or items which are highest in quantity but low in value and make up almost 5% of the total revenue.

Eg : Apsara pencil, pens, Extra long Eraser(Rubber), Colors, chart papers, etc.

Hence, the ABC analysis provided the data-driven solution and correctly identified the products that are financially critical to the business. This will help the business to carefully manage the inventory based on Category A, Category B and Category C, which ultimately reduces the chances of stock-out for the high demand items and over-stocking of the low demand items.

5.3 Trend Analysis :

The Trend analysis is done to understand and identify the patterns in data. Unlike predictive regression (which tries to forecast the future), trend analysis focuses on describing what actually happened. Hence, it's appropriate as it sticks to the facts available in the data. The total sales revenue and the item quantities on a monthly basis is analyzed to understand the business's performance. The below are some of the trends seen during the 3-months.

- The sales peaked in April which accounted for **₹3,07,755** this was followed by a decline in May (₹1,23,166) and June (₹85,277). This indicates a "back-to-school" rush.
- In April, the sales were almost entirely driven by two categories : **Books** and **Paper Goods**. And after April, the sales for these category items dropped as the customers had already made their first purchase.
- In the Stationary category which is low-value and high-volume items, the sales were lowest in April and then grew into June as it is not tied to the academic rush.

Hence, the Trend analysis justified the ABC analysis, and the bookstore must have enough "Category-A" items for the April peak and must avoid overstocking of these category items during the months of May and June.

6. Results and Findings

6.1 Monthly Transaction Patterns :

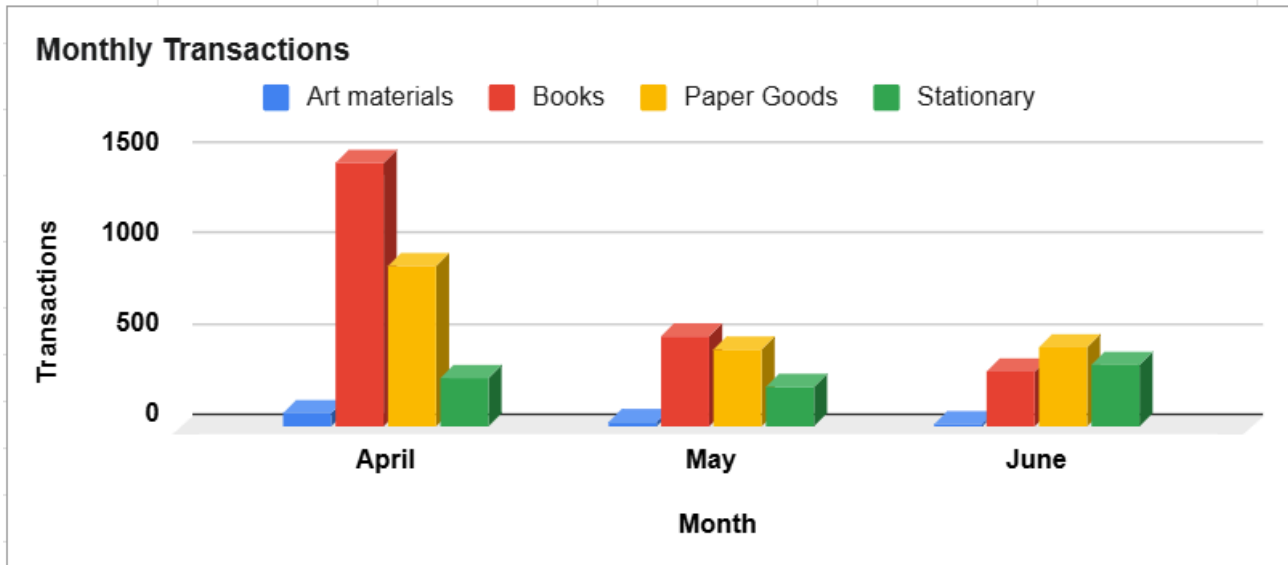


Chart-1 : Monthly Transactions for each category

As mentioned in the earlier sections, the **Books** were highest sold during the month of April due to the “back-to-school” rush and customers made their first academic purchase during this month. The demand for books then declined (from 1455 to 495 transactions) in May and again declined (from 495 to 306 transactions) in June. During the month of June the **Paper Goods** overtook Books as the academic term settled, this can be clearly seen in the Column chart above.

6.2 The Total Revenue of the Bookstore :

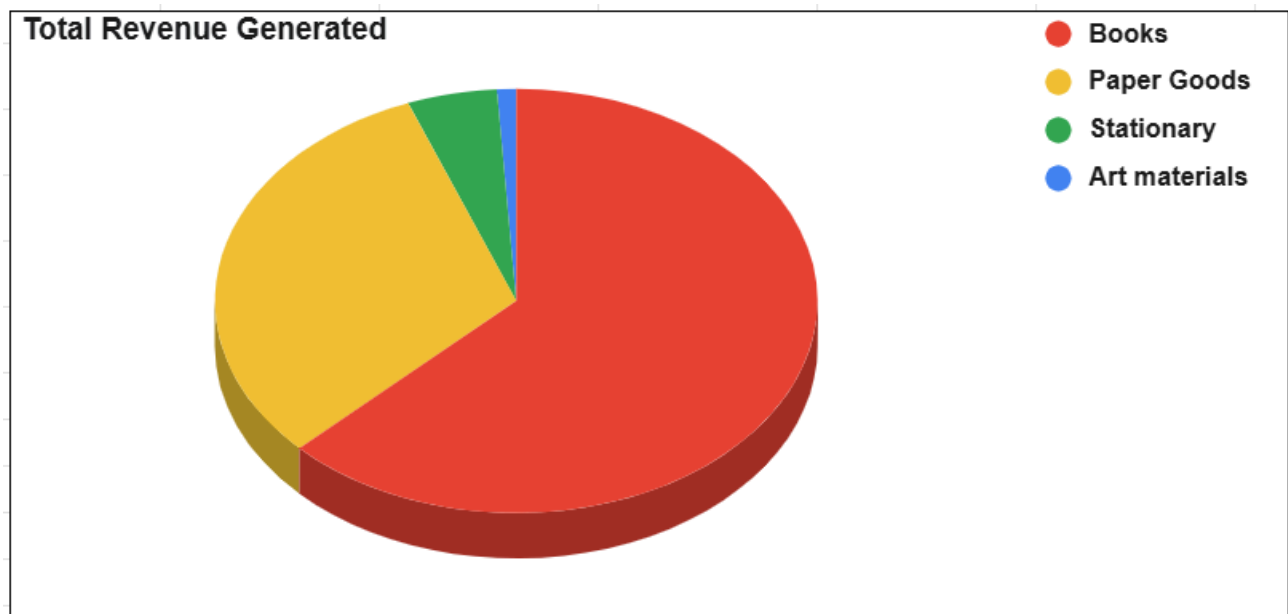


Chart-2 : Pie chart showing Category wise Total Revenue of 3-months

The Chart-2 (Pie chart) clearly shows that the “Books” category contributes Rs. 3,24,120 (62.8%) followed by “Paper Goods” category with Rs. 1,61,957 (31.4%), then “Stationary” category with Rs. 24,834 (4.8%) and at last “Art Materials” category with Rs. 5,287 (1%) of the total revenue during the 3-months. This also confirms the ABC analysis done in the previous section that category A items contribute the most in the total revenue of the business.

6.3 The Monthly Sales Trend by Category :

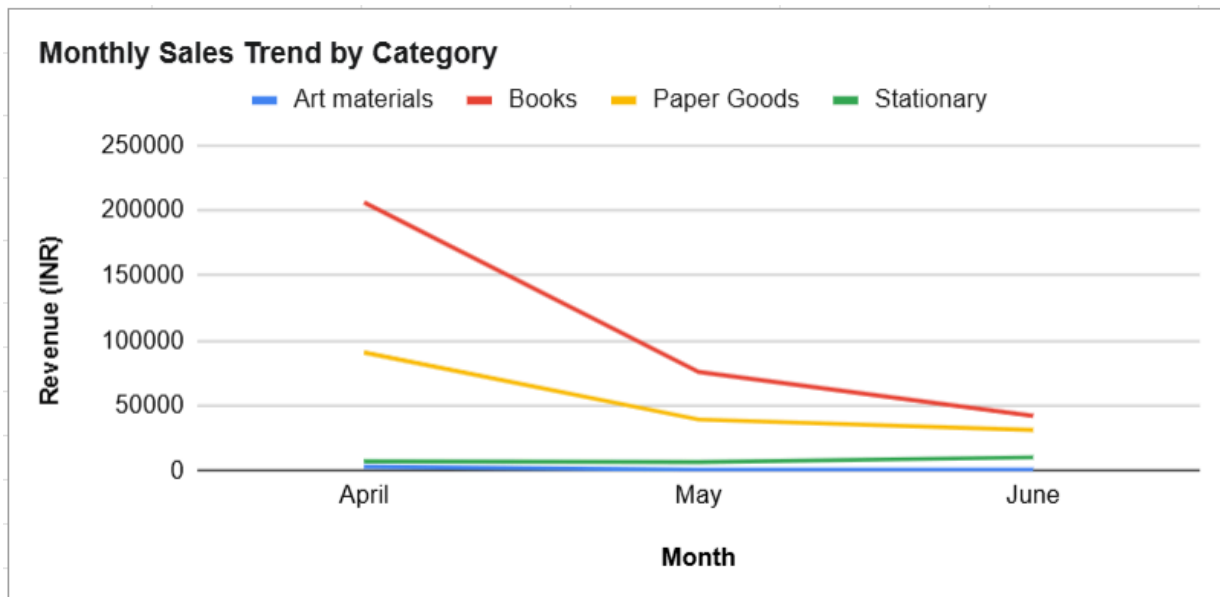


Chart-3 : Line Chart showing the Sales and Demand Trends

From the above line chart the sales and demand trends can easily seen, the most obvious insight is the massive drop in the **Books (red)** and **Paper Goods (yellow)** lines after April. This is the "back-to-school" rush in action. It proves that these categories are the seasonal drivers, responsible for the April peak, and then a sharp decline as the term settles. The **Stationary (green)** line was stable and even rose slightly in June, this confirms that "Stationary" items are not seasonal, they are consistent, acting as a steady traffic driver for the store. The **Art materials (orange)** line is the lowest and flattest of all, it's not a significant driver of revenue.